CS383 · SOFTWARE ENGINEERING

Lecture 1 Introduction (I)

Midterm summary

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01 Why do we care about Software Engineering?

Because a small error in a program can cost a huge amount of money. The slides mention three incidents:

Incident	What happened	Cause
Knight Capital	A loss of 440 million dollars in 30 minutes.	A programming error (bug).
Mars Climate Orbiter	A spacecraft worth 125 million dollars was lost.	The units were not converted from imperial to metric.
Colonial Pipeline	The fuel pipeline stopped, data was stolen, and a 5 million dollar ransom was paid.	Stolen login credentials, and no multifactor authentication.

02 Software costs

- The cost of software is often higher than the cost of hardware.
- Maintaining a program costs more than developing it.

Item	Percentage
Development costs	60%
Testing costs	40%

Remember

Maintenance and **Evolution** cost more than development and testing together.

03 Basic definitions (FAQ)

Question	Answer
What is **Software**?	A computer program together with its associated documentation.
What is **Software Engineering**?	A discipline concerned with all aspects of producing **cost-effective** and **trustworthy** software.
What are the attributes of **Good software**?	It meets the functional and non-functional requirements efficiently, and it is **maintainable**, **dependable** and **usable**.
What is the best development method?	It depends on the type of system. Games are built with prototypes, and critical systems need precise specifications and documentation.

The four fundamental activities

1. Software specification
2. Software development (design + implementation)
3. Software verification & validation
4. Software evolution

System engineering vs Software engineering

Term	Meaning
System engineering	More general. It covers hardware, software and process.
Software engineering	More specific. It focuses only on developing software.

04 Types of Software products

Type	Meaning	Examples
Generic products	A stand-alone system sold to any customer on the market.	Microsoft Office, CAD software, a dental clinic appointment system.
Customized products	A system tailored to one specific customer according to their request.	Embedded control systems, air traffic control, traffic monitoring.

05 Challenges and importance of SE

Challenges (Motivation)

Building software is hard because we need to know how to:

- Manage large programs.
- Build **scalable** programs.
- Deal with the high cost of development and maintenance.
- Deal with the changing nature of software.

- Produce programs of better quality.

Why is SE important?

Benefit	Explanation
Reduces complexity	It splits a big problem into small problems that are solved and tested separately.
Minimizes software cost	It lowers the cost of production.
Decreases development time	Working with a plan saves time.
Handling big projects	Big projects need planning, direction, testing and maintenance.
Reliable software	Testing and maintenance give confidence in the program.
Effectiveness	Following standards raises effectiveness.

06 Issues that affect SE

Issue	Meaning
Heterogeneity	Systems must work as distributed systems across networks that include different devices and mobile phones.
Business and social change	Business and society change quickly, so we must update programs and build new ones quickly.
Security and trust	We must block unauthorized people and protect information.
Scale	Programs are built at different sizes, from small embedded systems to cloud systems that serve the whole world.

07 Application types

There are many types of systems, and no single method suits all of them. This is called **Software engineering diversity**.

The choice of methods and tools depends on four things:

1. The type of application being developed
2. The requirements of the customer
3. The background of the development team
4. Tools used for development

Appeared in the midterm

An MCQ question: what determines the methods and tools that are used? The correct answer was:

> Application type, customer requirements, team background, and tools

08 Common mistakes

Mistake	Correct
Saying software is only the code	It is the code together with its associated documentation.
Saying development is the most expensive phase	Maintenance and evolution cost more.
Swapping Generic and Customized	**Generic** is for any customer. **Customized** is for one specific customer.
Saying there is one development method that suits all systems	There is not. It depends on the application type, the customer, the team and the tools.

09 Test yourself

1. Define Software Engineering.
2. List three attributes of Good software.
3. Which type is Microsoft Office: Generic or Customized?
4. What is the difference between System engineering and Software engineering?
5. Which issue is about protecting information and blocking unauthorized people?

Answers

#	Answer
1	A discipline concerned with all aspects of producing cost-effective and trustworthy software.
2	**Maintainable**, **Dependable**, **Usable**.
3	**Generic product**, because it is sold to any customer.
4	The first is more general and covers hardware, software and process. The second is about software only.
5	**Security and trust**.